

Dr. Rajesh Akula

Assistant Professor

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I am an Assistant Professor specializing in thermal management and intelligent control of lithium-ion battery systems. My research focuses on developing experimentally validated and data-driven cooling strategies for enhancing safety, performance, and energy efficiency in electric vehicles and energy storage systems.

Research Areas:

- i. Thermal Management and safety of Lithium-ion Batteries and Electronics
- ii. Phase Change Material based heat sinks
- iii. Machine Learning for Thermal and Energy Systems

Education

- i. M.S and Ph.D. (direct) 2017-2022: Department of Mechanical Engineering, IIT Madras, India.
Thesis title (July 2022): Thermal management of electronics and Li-ion batteries using phase change material-based heat sinks.
- ii. Bachelor of technology (2012-2016): RGUKT Nuzvid, Andhra Pradesh, India; CGPA - 8.92.

Professional Experience:

- i. Assistant Professor, IEST (2025 August - Present):
Developing intelligent thermal management strategies integrating machine learning and physics-based modeling for lithium-ion battery systems
- ii. Postdoc-II, University of Toronto (2024-2025):
Worked in industry-collaborative project with Ford Motor Company on thermal characterization, life assessment, and advanced cooling of Li-ion batteries
- iii. Postdoc-1, IIT Bombay (2023-2024):
Developed machine learning-based smart thermal management frameworks for Li-ion battery systems

Research Highlights

- i. Developed machine learning-enabled thermal management strategies for Li-ion batteries
- ii. Designed PCM-based cooling systems validated through experiments and simulations
- iii. Published in leading journals such as Applied Energy, Journal of Energy Storage, and Applied Thermal Engineering

Knowledge Dissemination

Currently contributing to an invited book chapter titled "Deep Learning in Heat Transfer and Thermal Control" for the book "**Handbook on Intelligent Thermal Systems**" to be published by Springer Nature.

Publications

Publication Summary:

- i. 13 international journals (Q1 and Q2)
- ii. 8 international conferences
- iii. 480+ citations

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Top 5 Selected Publications:

- i. **R. Akula**, C. Balaji*, "Thermal management of 18650 li-ion battery using novel fins-PCM- eg composite heat sinks", *Applied Energy* 316 (2022) 119048
<https://doi.org/10.1016/j.apenergy.2022.119048>
- ii. **R Akula**, L Kumar*, "Machine learning-based prediction of coolant velocity profiles for thermal management and energy minimization across drive cycles", *Journal of Energy Storage* 113, 115585
<https://doi.org/10.1016/j.est.2025.115585>
- iii. **R. Akula**, A. Minnikanti, C. Balaji*, "Pin fin-PCM composite heat sink solution for thermal management of cylindircal Li-ion battery", *Applied Thermal Engineering* 248 (2024) 123146
<https://doi.org/10.1016/j.applthermaleng.2024.123146>
- iv. Z.A. Belwadi, **R. Akula***, "Design of a PCM-based thermal management system for cylindrical Li-ion battery using topology optimization", *Applied Thermal Engineering* 270 (2025) 126143.
<https://doi.org/10.1016/j.applthermaleng.2025.126143>
- v. **R. Akula**, A. Gopinath, S. Rangarajan, C. Balaji*, "Experimental and numerical studies on heat transfer from a PCM-based heat sink with baffles", *International Journal of Thermal Sciences* 159 (2020) 106525
<https://doi.org/10.1016/j.ijthermalsci.2020.106525>

Intellectual Property

Patent (Published ; Positive First Examination Report - FER):

Title: SMART THERMAL MANAGEMENT SYSTEM FOR LITHIUM-ION BATTERY MODULES

Patent Application No.: 202521082797

Peer Recognition

- i. Secured All India Rank of 1549 out of 190648 candidates in GATE-2017 with 99.1875 percentile (754 gate score), organized by IIT Roorkee.
- ii. Awarded postdoctoral equivalent fellowship by IIT Madras for submitting the final thesis within 60 months from the date of joining, in the year 2022.

Research Impact

Work cited over 480 times, contributing to advancements in thermal management and safety of lithium-ion battery systems.